



OAKLAND COMMUNITY COLLEGE*

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2040 FINAL

NO FLASH DRIVES ARE ALLOWED

IF A FLASH DRIVE IS FOUND IN THE COMPUTER A ZERO GRADE WILL BE ISSUED.

SAVE BEFORE DOWNLOADING.

IF THE SYSTEM FAILS AND YOU DID NOT SAVE, YOU MAY LOSE YOUR WORK AND WILL HAVE TO REDO.

ALLOTTED TIME:

FOUR (4) HOURS FROM THE COURSE SCHEDULED START TIME.

The following problems use Studio 5000, Studio 5000 Emulate, and FactoryTalk View SE Client

REMEMBER TO PRINT TO OneNote

Final problem Solenoids must use the filename: **SOLENOIDS** (10 Points)

Final problem Washer Station must use the filename: **WASHER**. (20 Points)

Final problem Clamping Station must use the filename: **CLAMPS**. (20 Points)

The following problems use Siemens TIA

REMEMBER TO PRINT TO OneNote

Final problem Start-Stop Delayed must use the filename: **DELAYED**. (15 Points)

Final problem Spool must use the filename: **SPOOL**. (15 Points)

The following problems uses Studio 5000, Studio Emulate 5000 and FactoryTalk View ME

DO NOT PRINT, HMI WILL BE CHECKED ON THE COMPUTER

Final problem FactoryTalk View ME. (20 Points)

Must use the filename **ROB2040 FINAL HMI**.

Use Studio 5000 SOLENOIDS

SOLENOIDS Studio 5000

Remember to print to OneNote in the FINAL Tab as SOLENOIDS

PROGRAM NAME: SOLENOIDS

Create a program that when the Normally Open START pushbutton is pressed and released, two actions will occur:

- a timer begins elapsing.
- Solenoid No. 1 is energized.

When the timer has completed its 5 second timing cycle, Solenoid No. 2 is energized.

Pressing and releasing the Normally Closed STOP pushbutton de-energizes Solenoid No. 1 and Solenoid No. 2 and resets the timer.

INPUTS:

Address	Controller Tag	Description
		Momentary NC pushbutton stop
		Momentary NO pushbutton start

OUTPUTS:

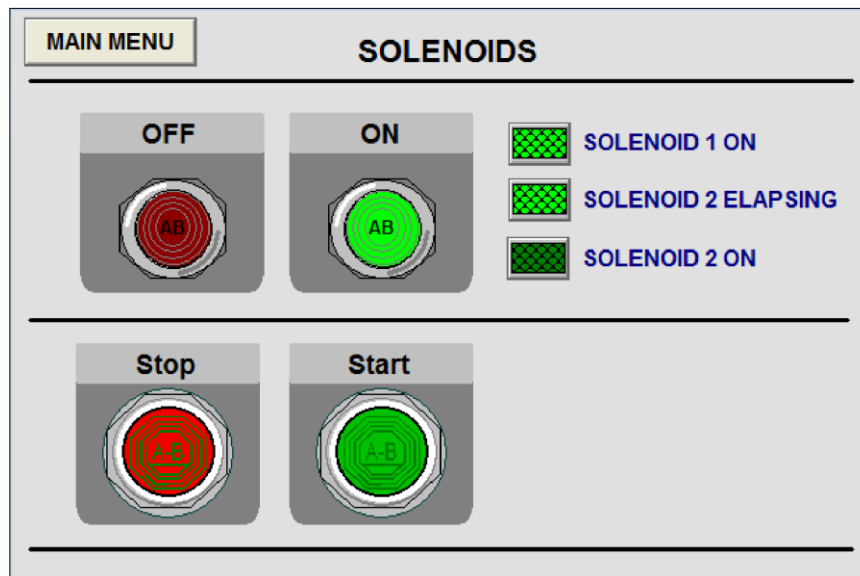
Address	Controller Tag	Description
		Solenoid No. 1
		Solenoid No. 2
		Power On Indicator
		Power Off Indicator

INTERNALS:

Param/Local Tag	Data Type	Description
		Timer for Solenoid 2

See the next page for HMI.

NOTES ON RUNNING THE ENVIRONMENT IN THE HMI:



Turn on the solenoids by placing the cursor over the START pushbutton and left click.

Following action should occur:

- the Power ON indicator is illuminated.
- Power OFF indicator is de-energized.
- SOLENOID 1 output is energized.
- the timer begins elapsing for Solenoid 2, shown by the indicator .
- after the time delay, SOLENOID 2 output is energized.

Move cursor over the STOP pushbutton and left click.

Following action should occur:

- the Power OFF indicator is illuminated.
- Power ON indicator is de-energized.
- SOLENOID 1 output is de-energized.
- SOLENOID 2 output is de-energized.

Remember to print to OneNote in the FINAL Tab as SOLENOIDS

WASHER STATION Studio 5000

Remember to print to OneNote in the FINAL Tab as WASHER STATION

PROGRAM NAME: WASHER

Create a program that when a pushbutton is pressed and held, two clamp solenoids will be energized. A normally open limit switch for each clamp will be set when the clamp is in position. When the two clamps are in position, a spray wash will clean the part for 10 seconds.

If the pushbutton is released before the end of the 10 second cycle,

- the spray wash valve *only* will turn off, the clamp solenoids will remain on,
- the washer timer will retain the current elapsed time.

When the pushbutton is again pressed and held,

- the spray wash valve will turn on,
 - and the washer timer will continue from the time value retained when the pushbutton was released.
-

At the end of the 10 second cycle,

- the spray wash valve only will turn off.

Releasing of the pushbutton at the end of the 10 second cycle will,

- release the clamps,

The released pushbutton at the completion of the timing cycle and the released clamps will,

- reset the timer,
 - increment a part counter.
-

The part counter preset is 500 and is used only to count parts washed, not to control the washing sequence.

The counter is reset when a normally open pushbutton is pressed.

continue

INPUTS:

Address	Controller Tag	Description
		Pushbutton Wash Cycle
		Limit Switch Clamp 1
		Limit Switch Clamp 2
		Counter Reset

OUTPUTS:

Address	Controller Tag	Description
		Clamp 1 Solenoid
		Clamp 2 Solenoid
		Spray Wash Valve

TIMER and COUNTER:

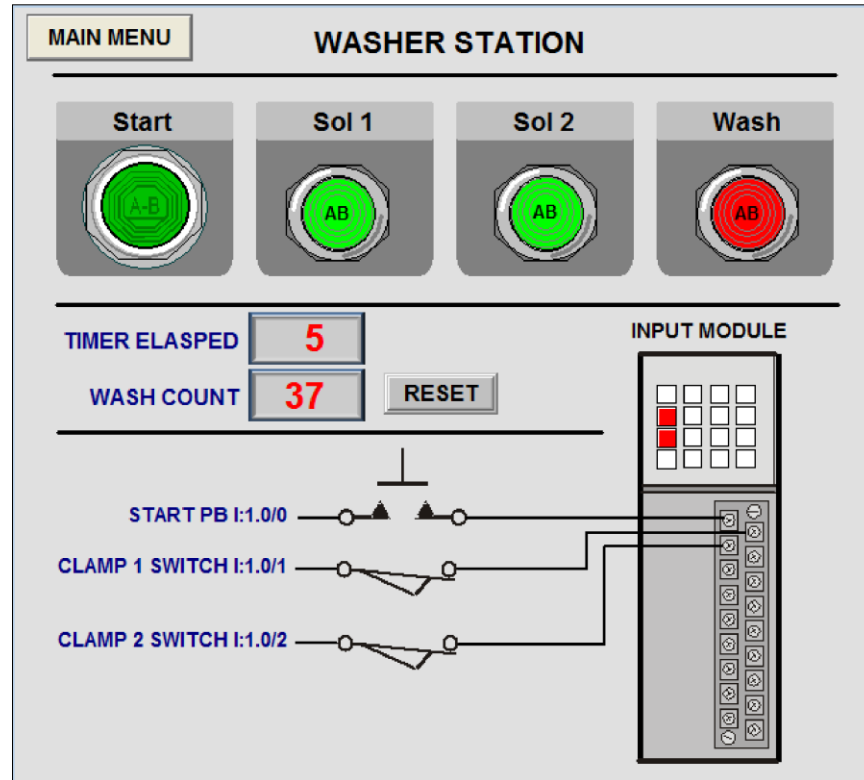
Use Timer Status Bits as needed for the program.

Param/Local Tag	Data Type	Description
		Spray Wash Timer
		Spray Wash Delay Complete

See the next page for HMI.

NOTES ON RUNNING THE ENVIRONMENT IN THE HMI:

Turn on the washer by placing the cursor over the START pushbutton, left click and hold.



The two outputs for the solenoid valves for the clamps should then turn on. After the inputs for the clamps are set, the spray wash should turn on.

If the START pushbutton is released before the end of the washing cycle, the WASH valve turns off, but, the SOLENOIDS remain energized and the clamps engaged. (Shown above)

When the START pushbutton is pressed again, the washing cycle continues at the current elapsed time.

When the washing cycle is complete, the WASH valve turns off and the clamps remain energized.

Releasing the START pushbutton de-energizes the SOLENOIDS and when the clamps are released the PART COUNT increments.

Remember to print to OneNote in the FINAL Tab as WASHER STATION

CLAMPING STATION - Studio 5000

Remember to print to OneNote in the FINAL Tab as CLAMPS

PROGRAM NAME: CLAMPS

Write a program where the PLC is interfaced to a machine controller. The machine controller operates the mechanisms that will place and remove the part. The PLC controls the clamps holding the part, and interfaces with the machine controller to coordinate the starting and completion of the machining process.

The machine places a new part and when it detected the PLC to engage the clamps. After all the clamps are closed, the PLC signals the machine to begin. At the completion of the machining process, the machine controller informs the PLC. The PLC opens the clamps and signals the machine to remove the part when all clamps are released.

-
- When the machine has placed the part, this will set the PART IN POSITION SIGNAL input signal. The PLC will engage all four hydraulic clamp SOLENOID VALVES.
 - Each clamp cylinder has normally an open LIMIT SWITCH to monitor its engaged position. When each clamp is engaged, a PILOT LIGHT for each clamp will be lit.

-
- With the part in position and when all four clamps are engaged, a START PROCESS output signal from the PLC will be set and sent to the machine controller to begin the machining cycle.

Hint: While the PART IN POSITION SIGNAL is set and all four clamps are engaged, the START PROCESS output turns ON.

-
- At the completion of the machining cycle, the machine will set PART PROCESS COMPLETE input signal for the PLC to release the clamps.
 - The PLC will de-energize the hydraulic SOLENOID VALVES. A spring return in the valve will open the clamps.
-



-
- Each clamps have a normally open limit switch to monitor when the cylinder has retracted to the disengaged position. When all clamps are disengaged, a REMOVE PART output from the PLC will be turned on to signal the machine controller to remove the part.

Hint: When the PART PROCESS COMPLETE input signal is set and all four clamps are disengaged, the REMOVE PART output turns ON.

- As the machine is performing the mechanical cycle to remove the part, this will cause the PART IN POSITION input to be reset .
 - After the part has been removed, the machine controller will then reset PART PROCESS COMPLETE input to the PLC .
-

All clamp limit switches are Normally Open

INPUTS:

Address	Controller Tag	Description
		Part in Position, Engage Clamps, Normally Open Proximity Switch
		Release clamps (Signal set from machine)
		Clamp No. 1 Disengaged Limit Switch
		Clamp No. 1 Engaged Limit Switch
		Clamp No. 2 Disengaged Limit Switch
		Clamp No. 2 Engaged Limit Switch
		Clamp No. 3 Disengaged Limit Switch

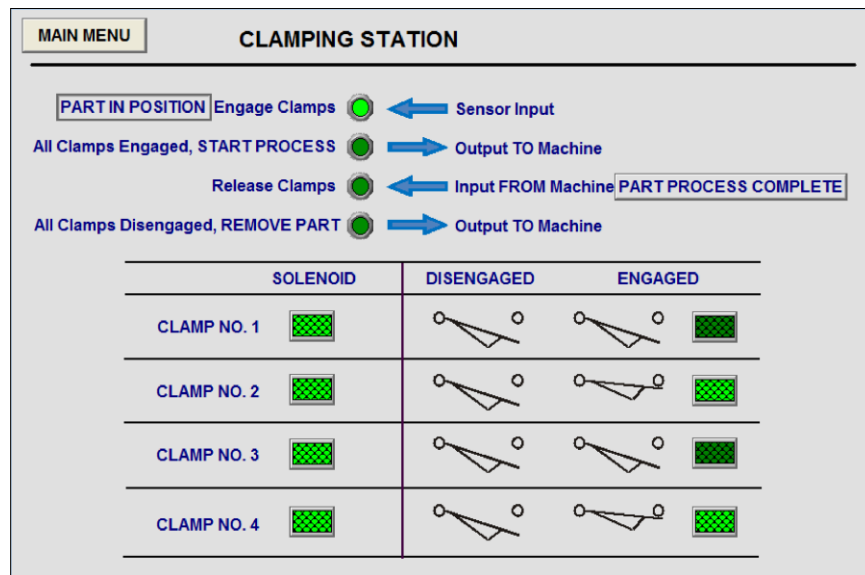
	Clamp No. 3 Engaged Limit Switch
	Clamp No. 4 Disengaged Limit Switch
	Clamp No. 4 Engaged Limit Switch

OUTPUTS:

Address	Controller Tag	Description
		All Clamps Engaged (Output to machine)
		All Clamps Disengaged (Output to machine)
		Clamp No. 1 Engage Solenoid Valve
		Clamp No. 1 Engaged Pilot Light
		Clamp No. 2 Engage Solenoid Valve
		Clamp No. 2 Engaged Pilot Light
		Clamp No. 3 Engage Solenoid Valve
		Clamp No. 3 Engaged Pilot Light
		Clamp No. 4 Engage Solenoid Valve
		Clamp No. 4 Engaged Pilot Light

See the next page for HMI.

NOTES ON RUNNING THE ENVIRONMENT IN THE HMI:



Place the cursor over the button for the PART IN POSITION input signal and left click.

If your program turns on the solenoid, the clamp's engaged limit switch input will be set by emulation environment.

When all clamps are engaged, your program should then turn On the START PROCESS output.

After the process has been started, place the cursor over the button for the PART PROCESS COMPLETE input signal and left click.

Your program should turn Off the START PROCESS output.

When your program turns off a solenoid, the emulation environment will set the clamp disengaged limit switch input.

When all clamps are disengaged, your program then should turn On the REMOVE PART output.

After your program sets the REMOVE PART output signal, the emulation environment will then turn Off the PART PROCESS COMPLETE and PART IN POSITION input signals.

Remember to print to OneNote in the FINAL Tab as CLAMPS

START-STOP DELAYED - Siemens TIA

Remember to print to OneNote in the FINAL Tab as DELAYED

PROGRAM NAME: DELAYED

This is a combination of the horn STARTUP TIMER lab and the TIMED CONVEYOR lab. The change from the TIMED CONVEYOR lab is the first output is now delayed to energize, turn on, after the horn has sounded.

Note: Tags are different than the lab print-outs.

Use delay to energize and delay to de-energize timers to create a delay Start/Stop Circuit. When the Start pushbutton is depressed, a horn will sound for five (5) seconds, then the horn will turn off and a first controlled motor will be energized. A second controlled motor will be energized after five (5) seconds after the starting of first controlled motor. The depressing of the Stop pushbutton will de-energize the first controlled motor after a five (5) second delay, and the second controlled motor will be de-energized after an additional five (5) second delay.

INPUTS:

Address	PLC Tag	Description
		Momentary N.O. START Pushbutton
		Momentary N.C. STOP Pushbutton

OUTPUTS:

Address	PLC Tag	Description
		Warning Horn
		Conveyor Motor No. 1
		Conveyor Motor No. 2

Timer tags on next page



TIMERS:

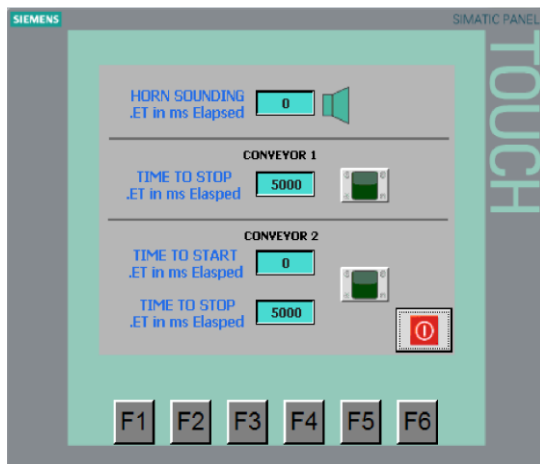
Use Timer Status Bits as needed for program.

Data Block Tag	Description
	Horn/Delay Energize Timer
	Delay Energize Timer for Motor No. 2
	Delay De-energize Timer for Motor No. 1
	Delay De-energize Timer for Motor No. 2

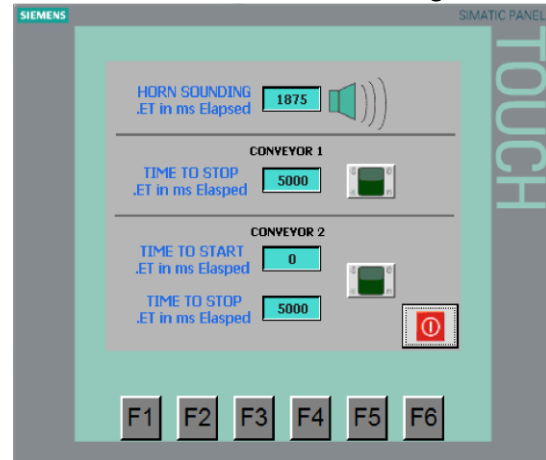
See next page for HMI and switches.

NOTES ON RUNNING THE ENVIRONMENT AND THE HMI:

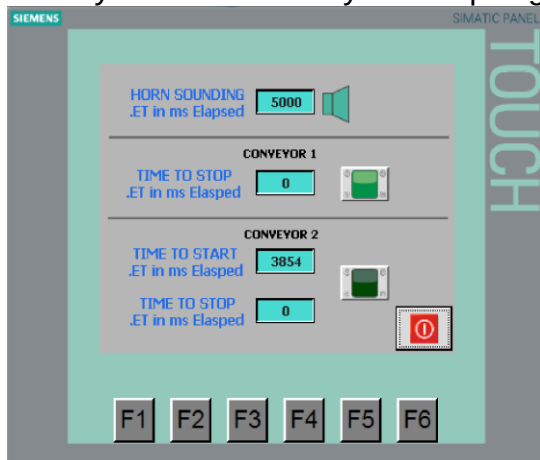
Initial Screen



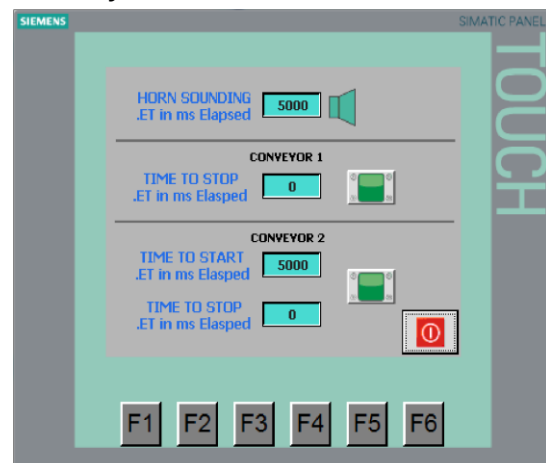
Start Presses - Horn Sounding



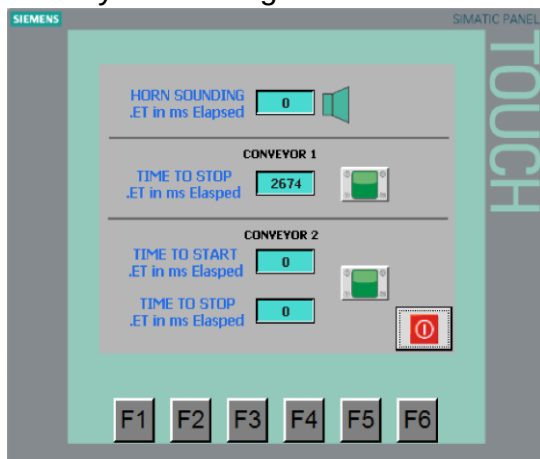
Conveyor 1 ON - Conveyor 2 Elapsing



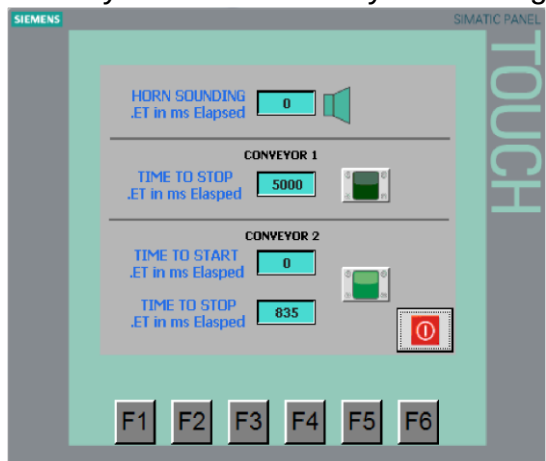
Conveyor 1 and 2 ON



Conveyor 1 Halting



Conveyor 1 OFF - Conveyor 2 Halting

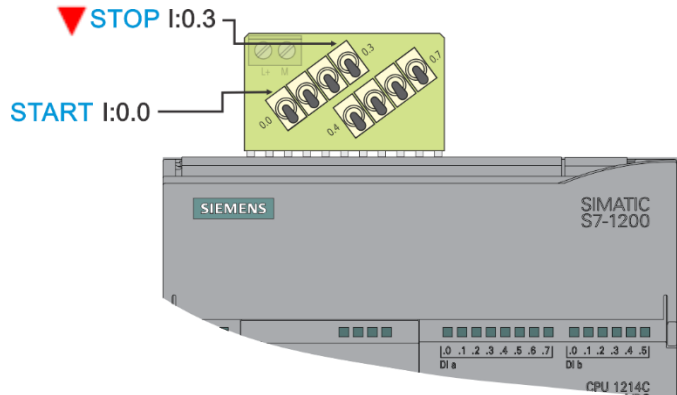


Operator Pressing the STOP pushbutton:

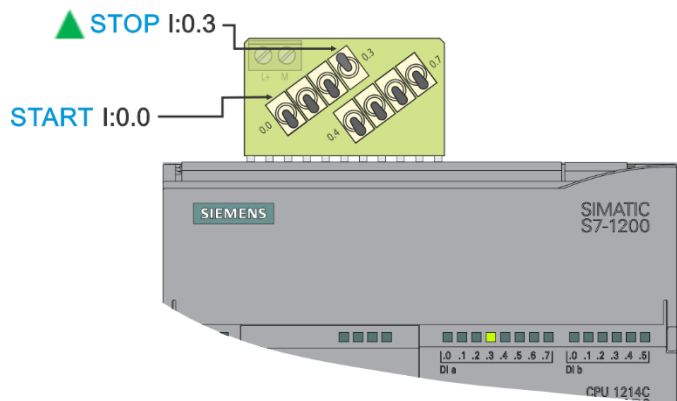
Push down ▼ the STOP switch to open.

When the first de-energize time delay has elapsed, the Motor No. 1 turns Off.

When the second de-energize time delay has elapsed, the Motor No. 2 turns Off.



Push up ▲ the STOP switch to close.



Remember to print to OneNote in the FINAL Tab as DELAYED

SPOOL - Siemens TIA

Remember to print to OneNote in the FINAL Tab as SPOOL

PROGRAM NAME: SPOOL

Note: the count preset values are set to low values for testing purposes.

Write a program to implement the process outlined. A company which assembles electronic kits needs a counter to count and control the number of resistors placed into each kit. The controller must stop the take-up spool at a predetermined amount of resistors (5). An operator will then cut the resistor strip and places it into the kit. Pressing and releasing the start pushbutton repeats the spooling/counting cycle. The circuit operates as follows:

Normally Open START/ Normally Closed STOP pushbutton station is used to turn the spool motor drive On and Off manually. Pressing and releasing the START pushbutton begins the spooling/counting cycle. Similar HMI START/STOP pushbuttons may also turn the spool motor drive On and Off.

A Normally Open thru-beam sensor counts the resistors as they pass by. The resistors will only be counted if the take-up spool output is on (set).

A Kit counter preset for 5 (the amount of resistors in each kit) will automatically stop the take-up spool when the accumulated count reaches 5.

If the take-up spool is halted by the STOP pushbutton or by the Grand Total counter (see: *Grand Total counter below*) before the Kit count of 5. The pressing of the START pushbutton will continue with the accumulated value of the Kit counter from the value at the time the spool count was stopped.

When the Kit counter has an accumulated value of 5 and the spool has stopped, the Kit counter will be reset automatically.

A Reset Normally Open Pushbutton or Normally Open HMI pushbutton may reset the Kit counter if the spool has stopped.

A second counter is provided to count the grand total used.

Grand Total Counter: When a count of 15 has been reached on the Grand Total counter, the spool will be turned off if running or prevented from being turned on. The operator will replace the roll of resistors and manually reset the grand total counter to allow operations to begin.

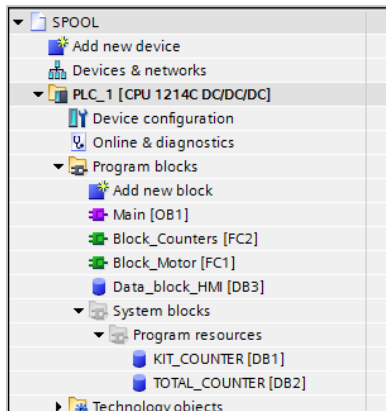
A Reset Normally Open Pushbutton or Normally Open HMI pushbutton may reset the Total counter if the spool has stopped.

continued



Project Structure:

The Main (OB1) only contains the unconditional calls to the Functions below.



Create the following Functions.

Name of Function: Block_Counters

Contains the counters and logic for the Kit and Grand Total.

Name of Function: Block_Motor

Contains the motor and its control logic.

INPUTS:

Address	PLC Tag	Description
		Momentary N.O. Pushbutton
		Momentary N. C. Pushbutton
		Thru-beam N. O. Sensor Detects Resistors
		Kit Counter RESET, Momentary N. O. Pushbutton
		Grand Total RESET, Momentary N.O. Pushbutton

OUTPUTS:

Address	PLC Tag	Description
		Take-up Spool

COUNTERS :

Use Counter Status Bits as needed for the program.

Data Block Tag	Description
	Kit Parts Counter
	Grand Total Counter Per Spool

continued



The Data Block and Tags has been created and does not need to be modified.

Use the tags listed below in your program. The tags are used with instructions that are in series or parallel with the corresponding physical pushbuttons (switches) in the logic to provide HMI equivalent of the wired pushbuttons.

Name of Data Block: Data_block_HMI

DATA BLOCK Data_block_HMI:

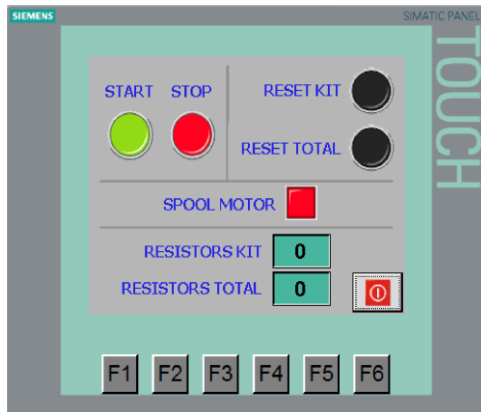
Data Block Tag	Data Type	Comments
	Boolean	Oper. Panel N.O. START Pushbutton
	Boolean	Oper. Panel N. C.STOP Pushbutton
	Boolean	Oper. Panel N.O. RESET KIT Pushbutton
	Boolean	Oper. Panel N.O. RESET TOTAL Pushbutton

See next page for HMI and switches.

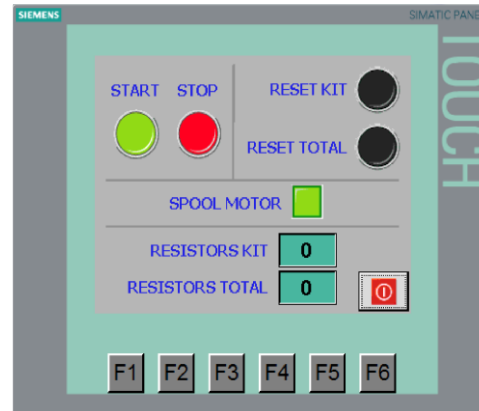


NOTES ON RUNNING THE ENVIRONMENT IN THE HMI:

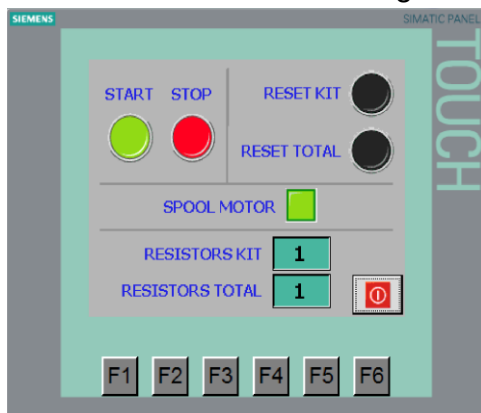
Initial HMI screen.



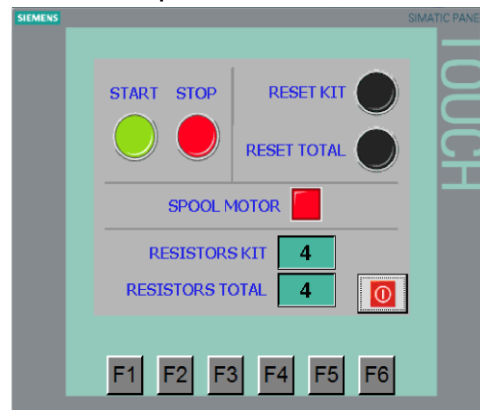
START pressed, MOTOR ON.



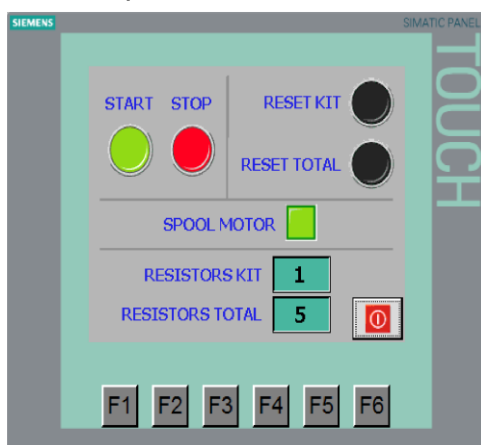
PHOTOELECTRIC counting.



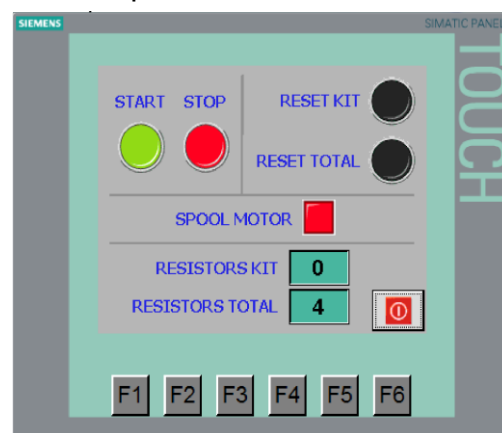
Count complete, MOTOR OFF.



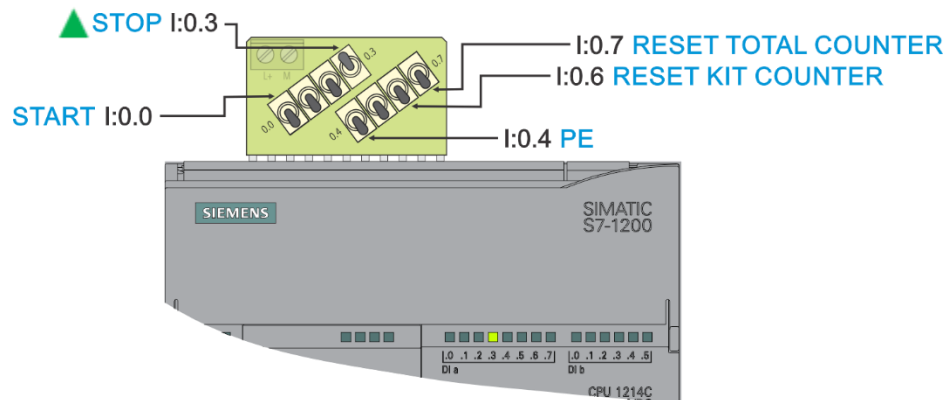
RESET pressed, Count zeroed.



START pressed, MOTOR ON, new



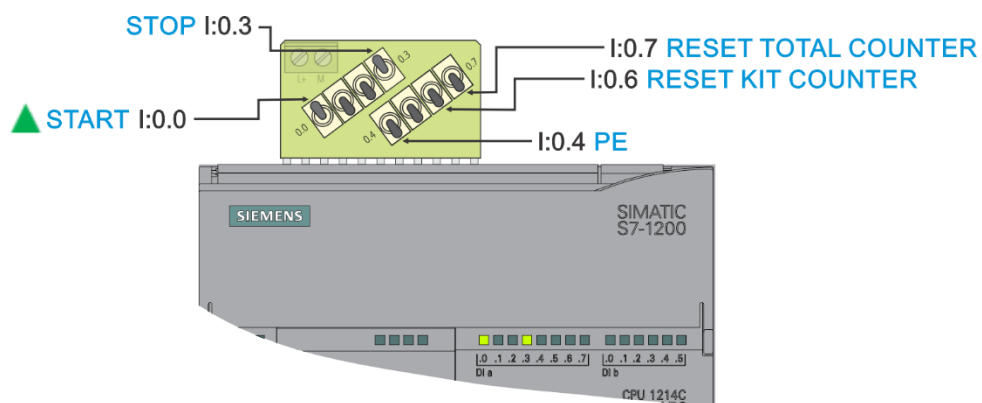
Initial switch settings, **push up ▲** the **STOP** switch to close.



THIS SECTION WILL START A CYCLE TO TEST THE RESET KIT COUNTER.

Operator Pressing the START pushbutton:

Push up ▲ the **START** pushbutton. *The take-up SPOOL turns ON.*

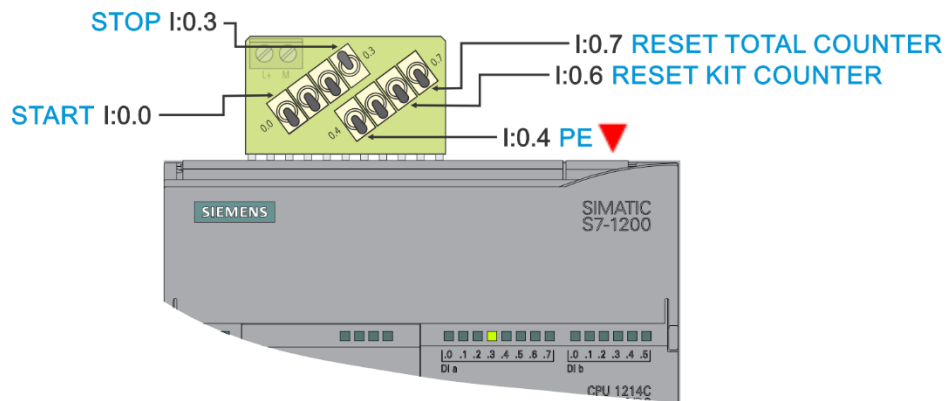
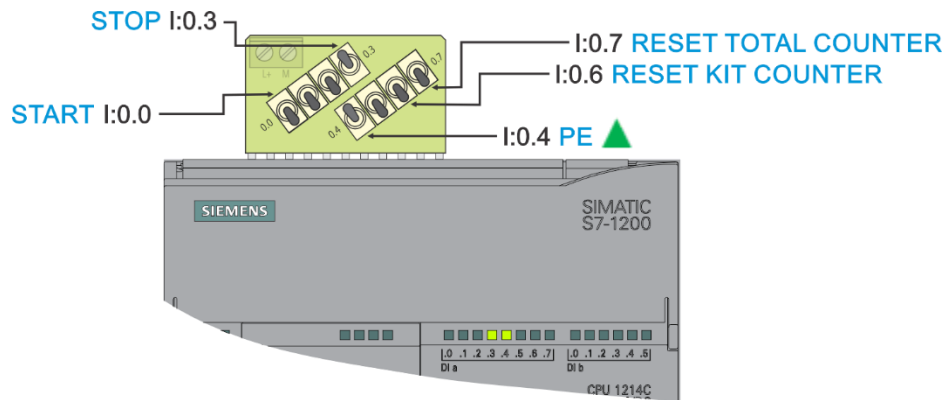


Push down ▼ the **START** pushbutton. ▼ **START I:0.0**



continued

As the Motor runs, toggle up and down PHOTOELECTRIC switch until the part count is reached.



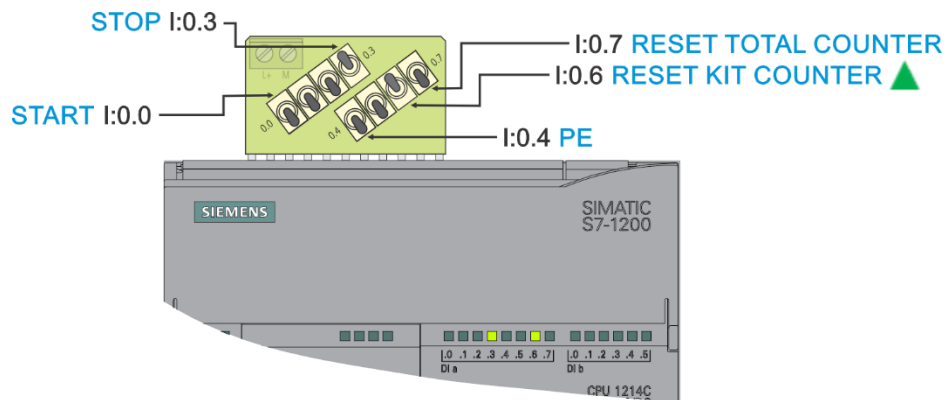
*When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.*

continued



Operator Pressing the RESET KIT pushbutton:

Push up ▲ the **RESET KIT** switch to close. **KIT COUNT** *returns to zero*..



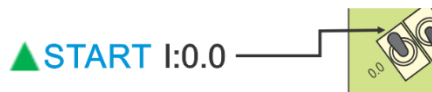
Push down ▼ the **RESET KIT** switch



THIS SECTION WILL START A CYCLE TO TEST THE RESET TOTAL COUNTER

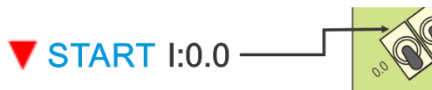
Operator Pressing the START pushbutton:

Push up ▲ the **START** pushbutton.



The take-up SPOOL turns ON

Push down ▼ the **START** pushbutton.



As the Motor runs, toggle up and down **PHOTOELECTRIC** switch until the part count is reached.

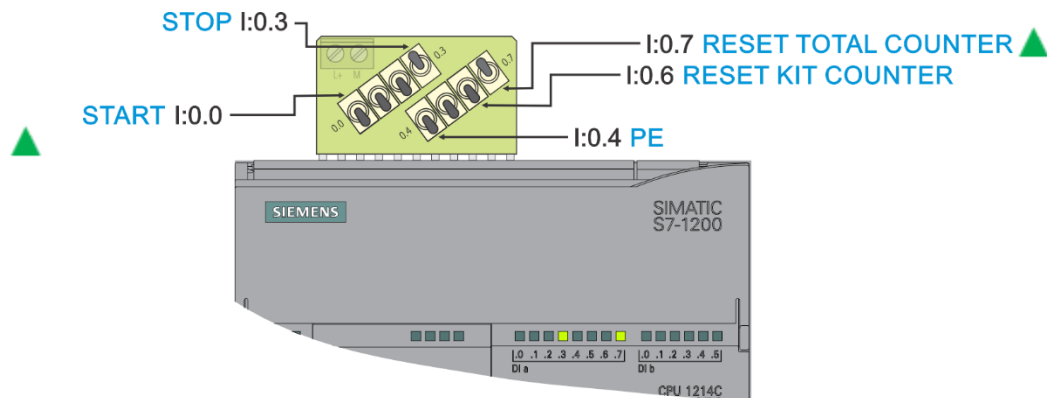


*When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.*

continued

Operator Pressing the RESET TOTAL pushbutton:

Push up ▲ the **RESET TOTAL** switch to close. **TOTAL COUNT** *returns to zero.*



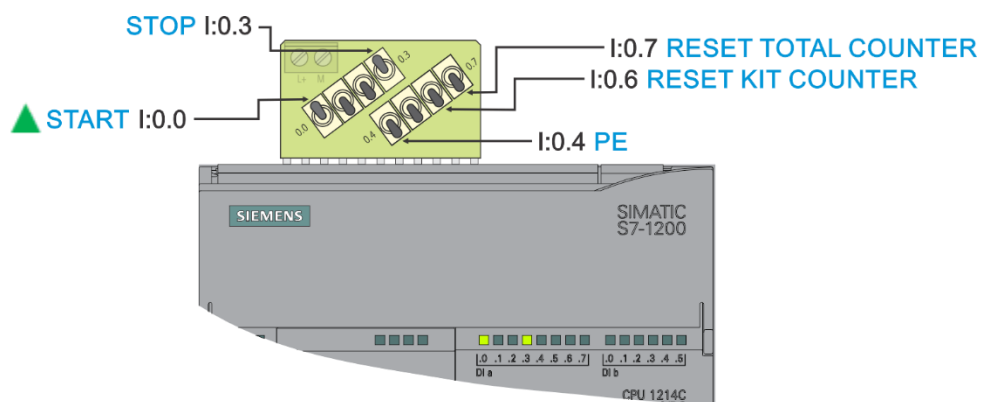
Push down ▼ the **RESET TOTAL** switch.



THIS SECTION WILL START A CYCLE TO TEST THE **STOP** pushbutton.

Operator Pressing the START pushbutton:

Push up ▲ the **START** pushbutton. *The take-up SPOOL turns ON.*



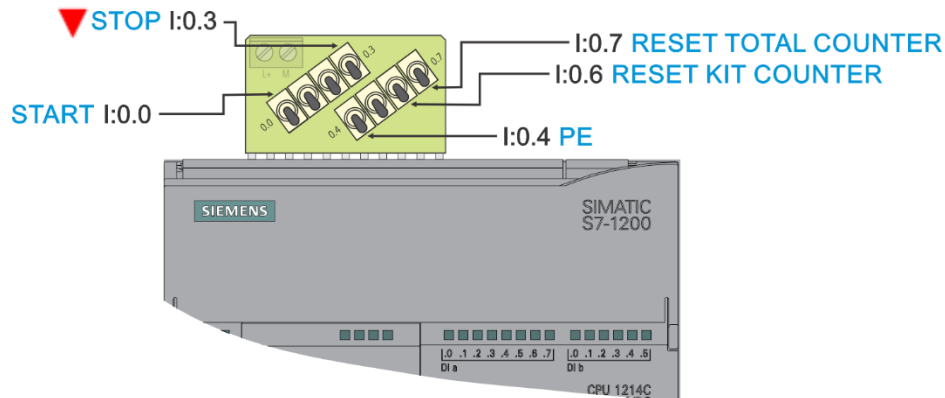
Push down ▼ the **START** pushbutton. ▼ **START** I:0.0



continued

Operator Pressing the STOP pushbutton:

Push down ▼ the **STOP** pushbutton. *The take-up SPOOL turns OFF.*



Push up ▲ the **STOP** pushbutton. ▲ STOP I:0.3



THIS SECTION WILL TEST THE HMI PUSHBUTTONS

Operator Pressing the HMI START pushbutton:



Press the **HMI START** pushbutton.

The take-up SPOOL turns ON.

As the Motor runs, toggle up and down **PHOTOELECTRIC** switch until the part count is reached.



*When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.*

continued

Operator Pressing the HMI RESET KIT pushbutton:



Press the **HMI RESET KIT** pushbutton.

TOTAL COUNT *returns to zero.*

Operator Pressing the HMI RESET TOTAL pushbutton:



Press the **HMI RESET TOTAL** pushbutton.

TOTAL COUNT *returns to zero.*

Operator Pressing the HMI START and then STOP pushbutton:



Press the **HMI START** pushbutton.

The take-up SPOOL turns ON.

Operator Pressing the HMI STOP pushbutton:



Press the **HMI STOP** pushbutton.

The take-up SPOOL turns OFF.

continued

THIS SECTION WILL TEST THE **RESETS WHILE THE MOTOR IS ON**

Operator Pressing the HMI START pushbutton:

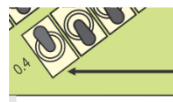


Press the **HMI START** pushbutton.

*The take-up **SPOOL** turns ON.*

Only toggle the switch for two counts.

As the Motor runs, toggle up and down **PHOTOELECTRIC** switch until the part count is reached.



I:0.4 PE ▲



I:0.4 PE ▼

Operator Pressing the HMI RESET KIT pushbutton:



Press the **HMI RESET KIT** pushbutton.

TOTAL COUNT

Operator Pressing the HMI RESET TOTAL pushbutton:



Press the **HMI RESET TOTAL** pushbutton.

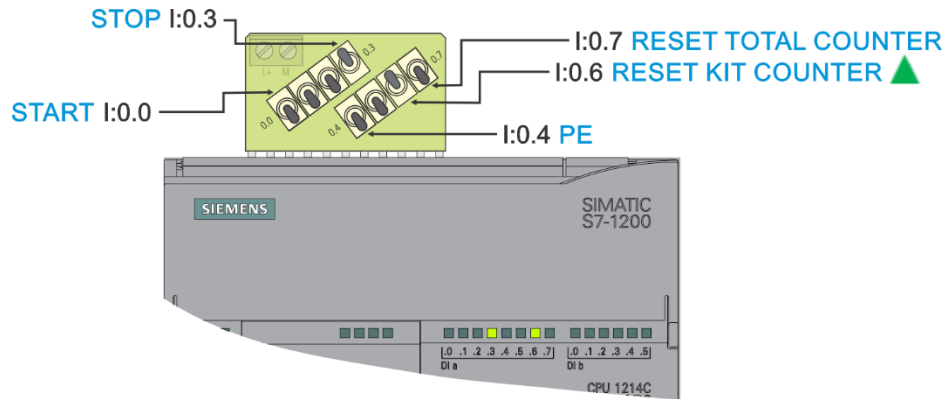
TOTAL COUNT *should not reset to zero.*

continued



Operator Pressing the RESET KIT pushbutton:

Push up ▲ the **RESET KIT** switch to close. **KIT COUNT** *should not reset to zero.*

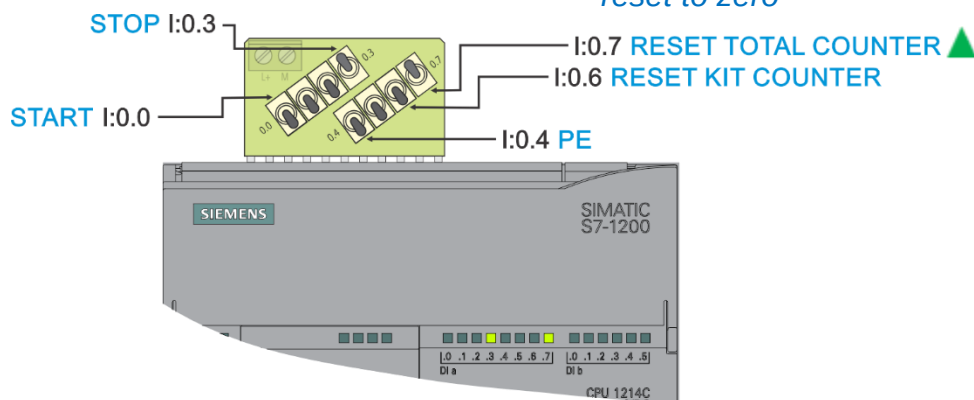


Push down ▼ the **RESET KIT** switch



Operator Pressing the RESET TOTAL pushbutton:

Push up ▲ the **RESET TOTAL** switch to close. **TOTAL COUNT** *should not reset to zero*



Push down ▼ the **RESET TOTAL** switch.



Press the **HMI STOP** pushbutton. *The take-up SPOOL turns OFF.*

Remember to print to OneNote in the FINAL Tab as SPOOL



FactoryTalk View ME

DO NOT print to OneNote, HMI will be checked on the computer

The HMI will use the SOLENOID program.

Open the Desktop Icon, **ROB 2040 Final HMI**, for requirements.



Icon may be different.

DO NOT print to OneNote, HMI will be checked on the computer